

CHAPTER 1: INTRODUCTION TO DATA COMMUNICATION AND NETWORK

1.Which OSI layer is responsible for reliable data communications?

A	Transport
B	Network
C	Application
D	Data Link

2.Basic concept of Data Communication except

A	Delivery
B	Application
C	Accuracy
D	Jitter

3.Which of the following is the examples of software programs used at the Application layer of the OSI Reference models for creating and transmitting messages.

A	Electronic Spreadsheets
B	Web browsers
C	Word processing
D	None of the answers

4. Name of layer 5 OSI Reference model is

A	Session
B	Application
C	Data Link
D	Network

5. _____ is the message that created at the Data-Link layer?

A	An TCP segment
B	A datagram
C	An IP packet
D	A frame

6. Which of the following architecture is used for the Internet web service

A	File Server System access
B	Internet access system
C	Client/server system
D	All of the above

7. Which standardization organization is responsible for the 802.3 standard ?

A	IEEE
B	ANSI
C	ISO
D	IAB

8. In _____ transmission, the channel capacity is shared by both communicating devices at all times

A	simplex
B	full-duplex
C	half-duplex
D	half-simplex

9. In a mesh topology, the relationship between the two devices is called ____

A	primary -to-peer
B	peer-to-primary
C	primary-to-secondary
D	peer-to-peer

CHAPTER 2 : PHYSICAL LAYER

1.What are the two basic types of communication systems?

A	Analog and digital
B	Optical and wireless
C	Analog and Optical
D	Wired and wireless

2.Disadvantages of analog communication except

A	Coding is not possible
B	Low bandwidth requirement
C	Noise affects the signal quality
D	Repeaters cannot be used between transmitters and receivers

3.Which of the following advantages of Analog communications

A	Coding is not possible
B	Noise affects the signal quality
C	FDM can be used
D	It is not suitable for the transmission of secret information

4 .The physical layer is concerned with _____

A	bit-by-bit delivery
B	process to process delivery
C	application to application delivery
D	port to port delivery

5.What are digital signals?

A	Continuous in nature
B	Discrete in nature and represent sequence of voltage pulses
C	Represented by continuous electromagnetic waves
D	Transferred between two or more points in a communication system

6.What is the effect of noise on analog communication?

A	Noise does not affect the signal quality.
B	Noise reduces the signal quality.
C	Noise can be separated from the signal.
D	Noise increases the signal quality.

7.What are the three different types of multiplexing?

A	TDM, FDM, CDM
B	TDD, FDM, CDM
C	TDM, FDM, SDM
D	FDD, TDD, TDM

8. Bits can be sent over guided and unguided media as analog signal by _____

A	digital modulation
B	amplitude modulation
C	frequency modulation
D	frequency modulation

9. The physical layer provides _____

A	mechanical specifications of electrical connectors and cables
B	electrical specification of transmission line signal level
C	specification for IR over optical fiber
D	all of the mentioned

CHAPTER 3 : DATA LAYER

1.The length of Media Access Control (MAC) address id of _____.

A	36 bits
B	48 bits
C	42 bits
D	96 bits

2.Types of error except

A	Single bit error
B	Multiple bits error
C	Burst error
D	Forward error

3.What is the data layer in the OSI Layered Model?

A	The first layer of the model.
B	The third layer of the model.
C	The second layer of the model.
D	The fourth layer of the model.

4.What are some functions of the data layer?

A	Providing a physical transmission facility.
B	Providing a direct interface to the application layer.
C	Regulating the flow of data so that slow receivers are not swamped by fast senders.
D	All of the above.

5.Which sub-layer of the data layer is responsible for controlling access to a shared media?

A	Error control
B	Framing
C	Media Access Control (MAC)
D	Addressing

6.What is the purpose of addressing in the data layer?

A	Regulating the flow of data
B	Encoding hardware addresses
C	Detecting and recovering data errors
D	Breaking up the bit stream into discrete frames

7.What is framing in the data layer?

A	Synchronizing machines for data transfer
B	Assembling bits into bytes
C	Encoding hardware addresses
D	Breaking up the bit stream into discrete frames

8.What are the two ways of error correction in the digital world?

A	Frontward Error Connection , Backward Error Connection
B	Backward Error Connection , Forward Error Connection
C	Forward Error Connection,Error Detecting Code
D	Forward Error Connection , Error Correcting Code

9.What is used as parity bits for error detection in Forward Error Connection?

A	Corrupted bits
B	Data frames
C	Redundant bits
D	Checksum bits

CHAPTER 4 : LOCAL AREA NETWORK

1. All these are types of VLANs except

A	Protocol VLAN
B	Static VLAN
C	Dynamic VLAN
D	Transport VLAN

2.A hierarchical LAN design includes the following three layers except

A	Access layer
B	Distribution layer
C	Media layer
D	Core layer

3.What is a LAN?

A	Local Area Netflix
B	Local Area Network
C	Llama Alpaca Negotiations
D	Local Arabian Network

4. What is a characteristic of a LAN?

A	Small geographical location such as a school or small office.
B	Really cheap to set up.
C	Uses less cables than a star.
D	All of the above

5.If all devices are connected to a central hub, then the topology is called :

A	Bus topology
B	Tree topology
C	Ring topology
D	Star topology

6.LAN works on which of the following Layers?

A	Datalink and Network
B	Physical and Datalink
C	Physical
D	Datalink

7.Which of the following is a small single site network?

A	LAN
B	MAN
C	DSL
D	WAN

8.Any combination of two or more topologies is a _____ .

A	Bus network topology
B	Hybrid network topology
C	Tree network topology
D	Ring network topology

9.Local area network architecture is categorized as:

A	Peer-to-peer,private -server
B	Peer-to-peer,public-server
C	peer-to-peer , client-server
D	Point-to-point,public-server

CHAPTER 5 : WIDE AREA NETWORK

1. _____ is a packet switched network designed for transmitting data over fixed lines.

A	ATM
B	ADSI
C	Frame relay
D	SONET

2. ATM is an acronym for _____ in data communication.

A	Asynchronous Teller Machine
B	Asynchronous Transfer Mode
C	Asynchronous Telecommunication Mode
D	Asynchronous Transmission Mode

3. Because ATM _____, which means that cells follow the same path, the cells do not usually arrive out of order.

A	is multiplexed
B	is a network
C	is asynchronous
D	uses virtual circuit routing

4. ATM's connection-oriented paradigm introduces significant delay for distant communication.

True
False

5. When setting up Frame Relay for point-to-point subinterfaces, which of the following must not be configured?

A	The Frame Relay encapsulation on the physical interface
B	The local DLCI on each subinterface
C	An IP address on the physical interface
D	The subinterface type as point-to-point

6. The Acme Corporation is implementing dial-up services to enable remote-office employees to connect to the local network. The company uses multiple routed protocols, needs authentication of users connecting to the network, and since some calls will be long distance, needs callback support. Which of the following protocols is the best choice for these remote services?

A	Frame Relay
B	Point-to-Point Protocol (PPP)
C	High-level Data Link Control (HDLC)
D	Transmission Control Protocol (TCP)

7. A public switching telephone network (PSTN) uses which switching technique?

A	Packet
B	Circuit
C	Message
D	None of these

8. Suppose that you have a customer who has a central HQ and six branch offices. They anticipate adding six more branches in the near future. They wish to implement a WAN technology that will allow the branches to economically connect to HQ and you have no free ports on the HQ router. Which of the following would you recommend?

A	Frame Relay
B	Point-to-Point Protocol (PPP)
C	High-level Data Link Control (HDLC)
D	Transmission Control Protocol (TCP)

9. What technology takes data and breaks them into packets and sends them over a network, sometimes using different routes for each packet?

A	Circuit switching
B	Network switching
C	Message switching
D	Packet switching

CHAPTER 6 : NETWORK LAYER

1. How many bits are in an IPv4 address ?

A	8
B	16
C	32
D	64

2. What is the subnet address for host 12.2.2.127/16 ?

A	12.2.0.0
B	12.2.2.0
C	12.2.2.112
D	12.2.2.100

3. Which of the following IP addresses for the network 124.16.32.55/28 can be assigned to hosts ?

A	124.16.32.45
B	124.16.32.48
C	124.16.32.63
D	124.16.32.62

4. Which of the following are not functions of the network layer ?

A	Addressing packets with an IP address
B	Delivery reliability
C	Routing
D	Encapsulation

5. What is the class for IP address 192.168.100.10/24 ?

A	class A
B	class B
C	class C
D	class D

6. A host can be identified by _____ while a program running on the host can be identified by _____.

A	An IP address; a port number
B	An IP address; a host address
C	A port number; an IP address
D	An IP address; a well-known port

7. Which one is NOT a valid mask?

A	255.255.255.254
B	255.250.0.0
C	255.255.224.0
D	All of these

8. If you have a network with 4 subnets, how many bits for you borrow from a Class C to have enough sub-network?

A	2
B	3
C	4
D	5

9. What is the purpose of Class D TCP/IP ?

A	Large network
B	Medium network
C	Reserved
D	Multicasting

CHAPTER 7: TRANSPORT & APPLICATION LAYER

1. The _____ fields is used for error detection.

A	Checksum
B	Urgent pointer
C	Sequence number
D	Acknowledgement number

2. The _____ fields is used to order packets of a message.

A	Checksum
B	Urgent pointer
C	Sequence number
D	Acknowledgement number

3. This protocol provides identification of source and destination port addresses.

A	Transmission Control Protocol (TCP)
B	User Datagram Protocol (UDP)
C	File Transfer Protocol (FTP)
D	Address Resolution Protocol (ARP)

4. Which layer in OSI model performs the process to represent encrypts and decrypts data.

A	Presentation
B	Network
C	Application
D	Physical

5. On which OSI layer do Pocket Filtering Firewalls work ?

A	At the application layer
B	At the transport layer
C	At the network layer
D	Both at the application and transport layer

6. Which protocol is used for E-Mail Traffic ?

A	IM
B	ARP
C	SMTP
D	DMS

7. Which protocol is used to load or retrieve files between the client and server communication?

A	Transmission Control Protocol (TCP)
B	User Datagram Protocol (UDP)
C	File Transfer Protocol (FTP)
D	Hypertext Transfer Protocol (HTTP)

8. Which of the following layers does the HTTP protocol work on?

A	Application layer
B	Data link layer
C	Network layer
D	None of these

9. What is the size of the destination port in the UDP protocol?

A	8 bits
B	16 bits
C	20 bits
D	32 bits

CHAPTER 8 : INTERNETWORKING

1. What is the TCP/IP protocol model?

A	A five layered model which provides guidelines for the development of universally compatible networking protocols.
B	A four layered model which provides guidelines for the development of universally compatible networking protocols.
C	A two layered model which provides guidelines for the development of universally compatible networking protocols.
D	A seven layered model which provides guidelines for the development of universally compatible networking protocols.

2. What is the 4 layer of TCP/IP model?

A	Datalink, Internet, Transport and Application
B	Physical, Datalink, Network and Application
C	Software, Application, Transport and Physical
D	Database, Switch, Transport and Hardware

3. What is a network?

A	A network is a system of interconnected computers or other communications devices that communicate with one another and share application and data.
B	A network is a software program used to manage email accounts.
C	A network is a device used to amplify signals for better reception.
D	A network is a single computer that can perform multiple tasks simultaneously.

4. What is a LAN Network?

A	A network that connects devices in a single location or a limited area under the same administrative control
B	A network that connects devices using satellite communication.
C	A network that connects multiple devices over a large geographical area.
D	A network that connects devices wirelessly.

5. Which of the following devices is used to connect different networks together?

A	Firewall
B	Switch
C	Router
D	Modem

6. What is the main difference between LAN and WAN topologies?

A	The speed of data transfer.
B	The type of cables used for network connectivity.
C	The number of connected devices.
D	The size of the network.

7. What is the function of routing protocols?

A	To provide security for network traffic.
B	Learn routing information from neighboring routers.
C	To manage network traffic and prevent congestion.
D	To connect multiple devices within a network.

8. What is the primary difference between a LAN and a WAN topology?

A	A LAN has faster network speeds than a WAN.
B	A LAN is a wired network, while a WAN is a wireless network.
C	A LAN is a network that is confined to a small geographic area, while a WAN spans a larger geographic area.
D	A LAN is used for personal communication, while a WAN is used for business communication.

9. What is the main difference between TCP and IP?

A	TCP is a connectionless protocol, while IP is a connection-oriented protocol.
B	TCP is responsible for error checking and retransmission of lost packets, while IP is responsible for fragmentation and reassembly of data packets.
C	TCP is responsible for breaking down data into packets, while IP is responsible for ensuring reliable delivery of those packets.
D	TCP is responsible for establishing a connection between two devices, while IP is responsible for routing data packets.

CHAPTER 9: NETWORKING OPERATION SYSTEM

1. What is the function of NOS?

A	All users can share resources and any type of data (file).
B	NOS allows all technology services.
C	NOS protects the data, information and hardware components from unauthorized users.
D	All of the above

2. Which one is the CORRECT one about the first generation of NOS?

A	Use the potential of multitasking, multithreading, memory management and context manipulation
B	Provides remote access to server/client machines.
C	Made networking commercially viable and were deployed on a wide range of products.
D	Make the migration path completely transparent to customers.

3. Which of the following is NOT a network operating system?

A	iOS
B	macOS
C	Linux
D	Windows Server

4. What is the role of the Networking Operating System (NOS)?

A	To provide graphic user interface for network administrators
B	To process data packets for faster network speeds
C	To facilitate physical connections between network devices
D	Serve as gatekeepers to data and applications on all manner of these networks.

5. What is the use of the Second Generation of Networking Operating System (NOS)?

A	To enable multitasking, multithreading, memory management and context manipulation
B	To improve network security and prevent cyber attacks
C	To facilitate wireless networking between devices
D	To provide enhanced graphical user interface for network administrators

6. What is the component of the Networking Operating System (NOS)?

A	Keyboard and Mouse
B	Speaker and Microphone
C	Software and Hardware
D	Printer and Monitor

7. What are types of Networking Operating System (NOS)?

A	Mobile NOS and Embedded NOS
B	Peer to Peer NOS and Client/server NOS
C	Cloud NOS and Virtualization NOS
D	Mainframe NOS and Supercomputer NOS

8. Which network operating system is used primarily in large enterprise networks?

A	macOS
B	NetWare
C	Linux
D	Windows Server

9. Which network operating system is free and open source?

A	NetWare
B	Windows Server
C	macOS
D	Linux

CHAPTER 10 : PRESENTATION LAYER OPTIMIZATION

1. Types of network monitoring tools?

A	Network Detection Tools
B	Spot Check Tools
C	Trending Tools
D	All of the above

2. Spot check network monitoring tools are designed for?

A	For troubleshooting
B	For network security
C	For network optimization
D	For network capacity planning

3. What is the function of the Simple Network Management Protocol (SNMP) Protocol?

A	Provides a mechanism for hackers to access and control your network
B	Provides a mechanism for users to access and modify network settings
C	Provides a mechanism for management to extract information from Management Information Base (MIB)
D	Provides a mechanism for viruses to spread across the network

4. What is the purpose of network security?

A	The process of optimizing network performance
B	The process of analyzing network traffic
C	The process of securing your network against physical threats
D	The defense you use to protect yourself against ever-increasing cybercrime.

5. What are the benefits of network security?

A	To limit the amount of data that can be stored on the network
B	To protect sensitive information
C	To loss or corruption of valuable data
D	To discontinued operational ability

6. What are the three key focuses of network security key?

A	Physical, Digital and Signature
B	Security, Biometric and Password
C	Protection, Detection and Response
D	Biometric, Protection and Digital

7. Protection is one of the key focuses of network security. What is the reason for having network security for protection?

A	To optimize network performance
B	To prevent network security intrusion
C	To prevent data breaches
D	To reduce network latency

8. Detection is one of the key focuses of network security. Detection refers to__

A	The process of securing your network against physical threats
B	The process of optimizing network performance
C	The process of analyzing network traffic
D	The resources that allow you to analyze network traffic and quickly identify problems before they can do harm

9. What is the effect of not having network security for your business?

A	Not having network security can reduce the amount of maintenance required for a business
B	Not having network security has no effect on a business
C	Not having network security can improve the efficiency of a business
D	Not having network security can result in network breaches and malicious threats